

DATA STRUCTURES AND ALGORITHMS

Hello World

Introduction To Programming Languages

Data Types

Primitive

- Int
- Char
- Long
- Float
- Boolean
- Double

Non Primitive

- String
- Long
- Double
- Integer
- Boolean
- BigInt

Conditional Cases

- If-Else Statements,
- Ternary Operator
- Nested Conditions

Switch Cases

- Switch Statement Syntax
- Case Blocks, Default Case
- Fall-Through And Break

1. Basics Of Programming

Design Patterns

- Singleton
- Builder
- Bridge
- Command
- Iterator
- Factory
- Adapter
- Observer
- Proxy

Variables

- Declaration
- Scope
- Constants

Loops

- For
- While
- Do While

Arrays

- Multidimensional Arrays
- Dynamic Size Arrays

Basic Data Structures

- HashMaps
- Sets

OOPS

Pillars

- Abstraction
- Polymorphism
- Inheritance
- Encapsulation

Components

- Classes
- Objects
- Static Members
- Getters, Setters

Design Principles

- DRY
- KISS
- SOLID

Time & Space Complexities

- Constant
- Linear
- Logarithmic
- Quadratic
- Cubic

Number System

- Binary
- Decimal
- Conversion

Arrays

- One-Dimensional
- Multidimensional
- Dynamic Arrays

Linked Lists

- Singly Linked List
- Doubly Linked List
- Circular Linked List

2. Building Blocks Of DSA

Binary Operations

- Left Shifts
- Right Shifts
- Count Bits

Bit Operations

- AND
- OR
- NOT
- XOR

3. Linear Data Structures

Hash Maps

- Hash Functions
- Hash Tables
- Linear Probing
- Quadratic Probing
- Double Hashing

Stacks

- Normal Stack
- Two Stack
- Min Stack

Queues

- Normal Queues
- Queues With Stacks
- Priority Queues
- Double Ended Queues

Trees

Types

- Binary Tree
- AVL Tree

Search

- Breadth-First Search
- Depth-First Search

Transversals

- Pre-Order Traversal
- Post-Order Traversal
- In-Order Traversal

Heaps

- Features & Common Use

Graphs

- Directed Graphs
- Undirected Graphs

Tries

- Features & Use Cases

4. Non Linear Data Structures

Searching

- Linear Search
- Binary Search
- Ternary Search
- Jump Search
- Exponential Search

String Manipulations

- Reverse Strings
- Rotate Strings
- Remove Strings
- Insert Strings

5. Algorithms

Sorting

- Bubble Sort
- Insertion Sort
- Quick Sort
- Selection Sort
- Merge Sort
- Counting Sort
- Bucket Sort